Narrative Review

An update on community water fluoridation, part 2

Fluoride exposure and children's intelligence (IQ) scores—relevance of systematic reviews and meta-analyses to community water fluoridation

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ABSTRACT

Background. The authors provide an overview of the literature on the association between fluoride (F) exposure related to community water fluoridation (CWF) and children's intelligence (IQ). It also critically evaluates the limitations of the evidence base.

Types of Studies Reviewed. This review is grounded in assessments from 4 government reports and supplemented by means of 3 independent reports and 7 other meta-analyses and systematic reviews. Collectively, these resources offer an overview of the literature on the relationship between F exposure and IQ. In addition, a meta-analysis of the association between urinary F and IQ is reanalyzed for its relevance to CWF.

Results. The authors found there is no association between F concentrations relevant to CWF and IQ scores. Although researchers reported an association in endemic areas (ie, with F levels exceeding 1.5 mg/L), the design and conduct of these studies have numerous methodological weaknesses, including a limited number of high-quality studies; weak study designs; inadequate attention to accurately measuring F exposure, confounding variables, and outcomes; and the use of inappropriate statistical procedures. There is no established biological mechanism to explain F's effect on IQ. These limitations raise questions about the findings and preclude discussion of causal relationships. Furthermore, differences in water intake among countries should be considered when making cross-national comparisons.

Conclusions and Practical Implications. The evidence does not support a causal link between F in drinking water and children's IQ. Dental and public health care practitioners should use this evidence to inform, educate, and empower patients and the public about the safety and benefits of CWF.

Key Words. Fluoride; water fluoridation; childhood IQ; urinary fluoride; systematic review; meta-analysis; dose-response; dental fluorosis.

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The Environmental Protection Agency (EPA) sets fluoride (F) standards for drinking water quality. In 2016, several interested parties urged the EPA "to protect the public and susceptible subpopulations from the neurotoxic risks of fluoride by banning the addition of fluoridation chemicals to water."[1] The EPA determined that the petition did not establish that fluoridation chemicals presented an unreasonable risk to health. Therefore, it denied the petition, prompting a lawsuit in 2017.[2]

In August 2024, a US District Court judge in the Northern District of California ruled on the *Food & Water Watch, Inc., et al. v. United States Environmental Protection Agency, et al.*, no.

17-2162 (N.D. Cal. 2024) lawsuit.[3] The judge stated that, although he "does not conclude with certainty that fluoridated water is injurious to public health," there still was an "unreasonable risk" of harm in the form of lower IQ in children.[3] The judge did not ban fluoridated water but said that the EPA must take regulatory action to address the risk posed by means of community water fluoridation (CWF) in the United States.[3] The findings of fact relied to a large extent on the draft National Toxicology Program's (NTP) systematic review of literature, which was subsequently revised and published in 2024 (NTP Monograph).[4] The EPA has appealed the district court judge's decision. The final decision will not be known for some time. All major health care organizations continue to support CWF as a safe, cost-effective, and cost-saving public health intervention with broad population reach that transcends socioeconomic status.[5]

We summarized the key findings of the NTP Monograph concerning F exposure and neurodevelopment and cognition, the relevance of other reviews to CWF, and assessed the limitations of the evidence base. In addition, we compared mean IQ scores from fluoridated and nonfluoridated areas and reanalyzed the subsequently published NTP meta-analysis from Taylor and colleagues[6] on urinary F and IQ as well as the mean-effects dose-response. Our focus was on the relevance of the body of evidence for CWF (ie, 0.7 mg F/L).

METHODS

This research did not involve human data or participants and used publicly available data; therefore, an institutional review board assessment was not necessary, per the institutional policies of the affiliated organizations.

Our review of the F-IQ literature was grounded in the assessments from 4 reports and 3 reviews.[7-13] These key resources, supplemented by means of 7 additional meta-analyses and systematic reviews,[6,14-19] offered a comprehensive overview of the literature on the relationships between F exposure and children's IQ. Crucial statements from the NTP Monograph[4] were adapted to formulate key messages and a summary table was created by means of abstracting the main conclusions from other reports. Data were also extracted from 5 cohort studies[20-24] to compare mean IQ scores from fluoridated areas with those from nonfluoridated areas; this information is presented graphically. In addition, we discuss the most recent US study relevant to CWF policy.[25]

In their meta-analysis, Taylor and colleagues[6] followed 3 statistical procedures: (1) a mean-effects meta-analysis, (2) a dose-response mean-effects meta-analysis, and (3) a regression slope meta-analysis of 13 articles[21,27-38] to assess the association between urinary F and IQ scores. Data from individual studies were pooled using a random-effects model.[21,27-38] However, Taylor and colleagues[6] did not use all reported regression coefficients in the studies to estimate the pooled effect size in nonendemic areas (ie, <1.5 mg F/L); instead, they calculated 7 of 13 regression coefficients.[39] Therefore, we reanalyzed the regression slopes reported in the 13 articles[21,27-38] Taylor and colleagues[6] included to estimate the association between person-level urinary F exposure and children's IQ for its relevance to CWF. Data concerning the regression coefficient for every 1-mg/L change in urinary F were extracted from these studies.

We used a random-effects model to pool the data, focusing on relevance to CWF. We created 4 subgroups on the basis of the following exposure matrices: water in nonendemic F areas ($\approx \leq 1.5$ mg/L), water in endemic F areas ($\approx > 1.5$ mg/L), salt fluoridation, and coal-burning F exposure. We then reproduced Taylor and colleagues'[6] overall pooled estimate.[39] However, instead of following Taylor and colleagues'[6] approach and combining the multiple coefficients from the nonlinear regression models reported by Yu and colleagues[30] and Xia and colleagues[38] into a single coefficient, we treated these as separate coefficients in the subgroup meta-analysis. Although Feng and colleagues[33] found a positive association in nonendemic areas, a coefficient relevant to CWF was not available. We contrasted the pooled estimates and reported heterogeneity statistics. Statistical analyses were conducted using Cochrane Review Manager software, Version 5.4.1.[40]

ABBREVIATION KEY

CWF: Community water fluoridation.
EFSA: European Food Safety Authority.
EPA: Environmental Protection Agency.
F: Fluoride.
NASEM: National Academies of Sciences, Engineering, and Medicine.
NTP: National Toxicology Program.
WHO: World Health Organization.

RESULTS

Findings from the NTP Monograph

The Office of Health Assessment and Translation tool was used in the NTP Mongraph[4] to classify studies as either low- or high-quality on the basis of their potential for bias.[41] Of the 72 reviewed

studies[20,21,27-30,42-107] in which researchers assessed the association between F exposure and IQ in children, 19 (26%) were considered to have a low risk of bias.[21,27-30,42-55] Among these, 16 were low-risk-of-bias cross-sectional studies[27-30,43-52,54,55] and were regarded as high-quality evidence in assessing the confidence in linking F to IQ deficits.[4] However, Levy[108] has suggested these studies should be considered as only having relatively higher and lower risk of bias, because many of the studies with low risk of bias were only low in comparison with more flawed studies.

According to the NTP, only 1[52] of the 19 studies or articles with low risk of bias did not report an inverse association between estimated F exposure and IQ in children. Moreover, of the 53 studies[20,56-107] with high risk of bias, 46 found an inverse association between estimated F exposure and IQ in children, but citations were not provided in the report.[7] Researchers with the NTP[4] further found that higher estimated F exposures (eg, approximations of exposure to drinking water F concentrations exceeding the World Health Organization [WHO] guideline of 1.5 mg/L) were consistently associated with lower IQ in children. All studies were conducted outside the United States.

On the basis of the National Academies of Sciences, Engineering, and Medicine (NASEM) review,[7] the assessment in an earlier draft that F was a presumed neurodevelopmental hazard was removed from the NTP Monograph.[4] The Box contains the adapted key takeaway messages from the NTP Monograph to summarize the findings and highlights the limitations.

Findings from the agency reports, reviews, and meta-analyses

Table 1 provides a summary of the findings on F and children's IQ with regard to relevance to CWF. Several reviews have not found a causal link between CWF F concentration and neurocognitive effects.[7,8,10-12] Both the Health Canada Expert Panel and the European Food Safety Authority (EFSA) Scientific Committee agreed there is insufficient evidence to recommend a specific point of departure and health-based value for neurocognitive effects.[9,10]

Evaluation of evidence base

Overall, the evidence base on the association between F and IQ in children discussed in the NTP Monograph[4] is large but not robust, suggesting deficiencies in study design and execution (Figure 1). Most articles included in Taylor and colleagues'[6] meta-analysis (41/74) were not reported in MEDLINE-indexed peer-reviewed journals (Supplemental Material, available online) and relied on nonprobability sampling. Furthermore, in most studies (64/74), researchers used cross-sectional designs to investigate the association between F and IQ (Supplemental Material). Such a study design examines the population at only 1 point in time and cannot distinguish whether F exposure influences cognitive outcomes or underlying cognitive, behavioral, or social factors influence F exposure. In addition, high F exposure was defined as at least 1.5 mg F/L of water, which is more than double the recommended concentration in community water in the United States.

In the 3 highest-ranked studies[22,24,25] in Figure 1, researchers reported null or beneficial associations, with children in fluoridated communities performing as well as or better than their counterparts in nonfluoridated communities on IQ and cognitive assessments. Taken together, these results provide robust evidence CWF is not associated with adverse cognitive outcomes.[22,24,25]

Limitations of Cross-Sectional and Ecological Studies in Etiological Meta-Analysis

In general, the findings from cross-sectional and ecological studies are vulnerable to confounding and reverse causation and cannot be reliably combined to estimate cause-and-effect relationships[119,120] (M. B. Jané, MS, J. Heathers, PhD, D. R. Grimes, PhD, unpublished data, February 2025). In responding to the 2016 petition, the EPA determined that the F-IQ cross-sectional studies were substantially affected by means of antecedent-consequent bias (also known as the chicken-and-egg problem).[2] This means it cannot be determined whether the F exposure occurred before or after the IQ health effects because both are evaluated simultaneously.[2]

In cross-sectional F-IQ studies, it is possible that higher IQ leads to lower F exposure in endemic areas.[16] In China, India, and Pakistan, concerted communication efforts (eg, advisories, access to low F water, and public awareness campaigns) have been undertaken in endemic fluorosis areas to

Box. Key messages adapted from the NTP* Monograph[4] and a review of limitations regarding the association between F† and lower IQ relevant to community water fluoridation.

Findings

In the NTP Monograph, researchers reported that higher estimated F exposures were associated with lower IQ in children. These studies were conducted outside the United States, where nearly all water sources exceeded 1.5 mg/L of F (> twice the US recommended concentration of 0.7 mg/L).

In the NTP Monograph, it was not addressed whether the sole exposure to F added to drinking water in some countries (ie, fluoridation at 0.7 mg/L in the United States and Canada) was associated with a measurable effect on IQ.

In the NTP Monograph, researchers reported no evidence that F exposure had adverse effects on adult cognition.

In the human studies that were cited, researchers cannot explain how F exposure might affect IQ.

In the animal studies cited in the NTP Monograph, researchers did not provide insight into whether F exposure affects IQ.

The overall quality of the experimental animal studies is poor, and there are relatively few well-designed and well-performed studies at lower F exposure levels (ie, < 20 ppm, which is roughly equivalent to human exposure of < 4 ppm).

The NTP conducted an animal study to address this specific gap and found no evidence of an effect on memory or learning performances.

Limitations Noted in the NTP Monograph and Other Reviews

Many of the original articles were in non–MEDLINE-indexed journals and the researchers provided limited details on methodology.

In many studies, researchers did not provide sufficient information to assess whether children exposed to high and low F were comparable with respect to socioeconomic status.

Most studies lacked information regarding F exposure or had serious limitations in the exposure assessment measures.

In many studies, researchers relied on spot urinary F, which is not suitable for estimating a person's long-term F intake.

Many studies did not account for other exposures (other than F), such as exposure to arsenic, or iodine deficiency, which also affect children's IQ.

Children's IQ assessments are difficult to standardize and administer, which can introduce substantial errors in studies of F exposure and IQ. This was evident in the Maternal-Infant Research on Environmental Chemicals study, where IQ scores varied substantially among 6 cities, regardless of fluoridation status.

In many studies, researchers did not use appropriate statistical methods to analyze the data and report the findings.

Results of meta-analyses showing associations between F exposures and lower IQ scores relied on cross-sectional and ecological studies, which preclude assessment of cause-and-effect (ie, the chicken-and-egg problem), and had inaccurate estimates of effect sizes.

No consideration was given to the strong possibility of reverse causality in F exposure-IQ studies.

In many studies, researchers provided only group-level data, and, therefore, findings from such studies cannot be assumed to apply to people within a study.

* NTP: National Toxicology Program. † F: Fluoride.

reduce F exposure.[121-123] Taylor and colleagues[6] assumed exposure likely preceded the outcome in cross-sectional studies with low risk of bias, as only children who had lived in each community since birth or had dental fluorosis were included. Such an assumption overlooks the reverse causality phenomenon, when families with high literacy and resources, who are likely to have children

Table 1. Association between F* and children's IQ and its relevance to CWF† in reports, systematic reviews, and meta-analyses: 2018-2025.‡

REPORT OR STUDY	SCOPE AND METHODS	KEY FINDINGS ON IQ	COMMENTS ON ASSOCIATIONS OBSERVED AT EXPOSURE LEVELS RELEVANT TO CWF	RECOMMENDATIONS OR POLICY IMPLICATIONS
Governmental Agency Reports and Commissioned Reviews				
Guth and colleagues,[11] 2020	Narrative review of human and animal evidence	Noted that in most epidemiologic studies, researchers reported IQ associations, but highlighted serious methodological deficiencies (eg, confounding, coexposures, and reverse causality).	The results of the review did not support the presumption that F should be assessed as a human developmental neurotoxicant at the exposure levels in Europe.	Recommended high-quality animal and prospective epidemiologic studies.
National Academies of Sciences Engineering and Medicine,[7] 2021	Review of NTP§ draft monographs	Concluded that NTP had not provided a clear and convincing rationale for its "presumed hazard" conclusion in the draft documents; urged stronger methods and caution. Recommended avoiding impressionistic writing.	Advised the NTP to make it clear that the monograph cannot be used to draw conclusions regarding low F exposure concentrations (< 1.5 mg F/L), including those typically associated with drinking water fluoridation.	Directed NTP to clarify methodology and avoid unsupported claims.
Health Canada Expert Panel,[9] 2023	Expert panel review for drinking water values	Recognized IQ and neurocognitive effects as a candidate end point, but "not a sufficient basis at this time to recommend a point of departure or health-based value."	Evidence is uncertain at Canadian exposure levels.	Did not derive a neurocognitive-based limit; highlighted research gaps.
NTP,[4] 2024	Systematic review of human and animal studies	Found moderate confidence in the associations between higher F exposures (> 1.5 mg F/L) and lower IQ in children. Associations < 1.5 mg F/L unclear. Removed prior hazard statement in response to the National Academies of Sciences Engineering and Medicine review. Animal studies were uninformative.	Associations are consistent > 1.5 mg F/L. "This Monograph and Addendum do not address whether the sole exposure to fluoride added to drinking water in some countries (i.e., fluoridation, at 0.7 mg/L in the United States and Canada) is associated with a measurable effect on IQ."	No hazard classification or policy implications; descriptive only. Recommended more high-quality prospective studies at low levels of F.
Ministry of Health, New Zealand Government,[8] 2024	Review of F exposures relevant to New Zealand	Found IQ effects inconsistent at low to moderate exposures; signal mainly at > 1.5 mg F/L. Certainty is very low due to heterogeneity.	No high-quality evidence to suggest a causal link between F exposure at the levels used in Aotearoa New Zealand for CWF and significant harm to health.	Retained support for CWF; little evidence of harm at New Zealand fluoridation levels.
Academy of Medical Sciences,[13] 2025	Evidence synthesis report on the optimal concentration of water fluoridation in the United Kingdom	At suboptimal and optimal water F concentrations (< 1.2 mg/L) there is no evidence of an effect on neurodevelopment or cognition, as measured by IQ. Much of the evidence base for an adverse effect above 1.5 mg F/L for cognitive abilities is contested.	In a soon-to-be-published United Kingdom millenium cohort study,[26] researchers did not find evidence of an association between drinking-water F concentration and any of the cognitive ability scales in children.	Overall, water fluoridation programs are cost-effective but should be considered within the context in which the program is being implemented (eg, the population size and oral health status of the population).
European Food Safety Authority Scientific Committee,[10] 2025	Risk assessment of F for the European Union	Found evidence of IQ effects at > 1.5 mg F/L; evidence "not sufficiently consistent" < 1.5 mg/L.	1.5 mg F/L used as a benchmark; set protective intakes (eg, 3.3 mg/d for pregnant women).	The European Union legal limit (1.5 mg/L) may not be sufficiently protective for dental fluorosis; recommended reevaluation of the legal limit of F in drinking water.

* F: Fluoride. † CWF: Community water fluoridation. ‡ The internal dose of F at a given drinking water concentration is strongly influenced by means of water consumption patterns, which differ substantially across populations. Consequently, even at the same water F concentration (eg, 1.5 mg/L), populations in countries like China, where the water intake is higher, will receive a substantially greater internal dose. This distinction is critical when comparing studies conducted in different countries and climatic, occupational, or lifestyle contexts. § NTP: National Toxicology Program.

Table 1. Continued

REPORT OR STUDY	SCOPE AND METHODS	KEY FINDINGS ON IQ	COMMENTS ON ASSOCIATIONS OBSERVED AT EXPOSURE LEVELS RELEVANT TO CWF	RECOMMENDATIONS OR POLICY IMPLICATIONS
Systematic Reviews and Meta-Analyses				
Duan and colleagues,[19] 2018	Dose-response meta-analysis (mostly high-exposure areas)	Found lower IQ with higher F. The dose-response meta-analysis revealed a nonlinear pattern in both relative and absolute concentration of F in water. The relationship between F intake and intelligence levels was most obvious above an absolute F concentration of approximately 4 mg F/L. Socioeconomic status, a major confounding variable, was not provided in all original articles.	Limited relevance to CWF; most studies are cross-sectional, with high exposure in endemic areas. Most studies in the control group had exposure levels typically at approximately 1 mg F/L.	Recommended improvements to water quality and reducing the F levels in drinking water in areas with high F content in the water.
Miranda and colleagues,[17] 2021	Systematic review and meta-analysis	Found an association between F and reduced IQ >1.5 mg F/L, characterized by high heterogeneity; however, the evidence of certainty for this association is very low. Causality cannot be established.	Exposure to low or adequate water F levels was not associated with IQ deficits.	Recommended more studies.
Gopu and colleagues,[18] 2022	Systematic review	Results of most cross-sectional studies revealed negative associations, and longitudinal findings are mixed. Researchers in only 1 of 5 studies graded as excellent quality found a negative association between F exposure and cognitive outcomes, whereas in 30 of 34 poor and fair-quality studies researchers reported a negative association.	Negative associations are more consistent at ≥ 2 mg F/L. High-quality studies are less likely to find associations; confounding is a common issue.	Recommended more longitudinal studies with high methodological quality.
Kumar and colleagues,[15] 2023	Review or meta-analysis, focus was on nonendemic exposures relevant to CWF	Did not find an association <1.5 mg F/L	No association between F and IQ at F concentrations relevant to fluoridated communities.	Recommended interventional and longitudinal studies using robust methods.
Veneri and colleagues,[16] 2023	Systematic review and dose-response meta-analysis	Found an overall inverse association between F and IQ; the effect was strongest in studies with high risk of bias; no adverse effect in the only study with low risk of bias.	Suggested effects even at low levels (>0.7 mg F/L), but uncertain. However, a significant role of residual confounding could not be ruled out.	Recommended more prospective studies with low risk of bias.
Taher and colleagues,[14] 2024	Systematic review	Found evidence that F exposure reduces IQ levels in children, although there is some uncertainty in the causal association.	Associations are consistent at >1.5 mg F/L.	A point of departure of 1.56 mg F/L for moderate dental fluorosis may be preferred as a starting point for setting a health-based value for F in drinking water.
Taylor and colleagues,[6] 2025	Review or meta-analysis supplementing the NTP systematic review[4]	Found inverse associations and a dose-response association between F measurements in urine and drinking water and children's IQ.	Associations are consistent at >1.5 mg F/L.	These results may inform future comprehensive public health risk–benefit assessments of F.

with high IQs, tend to avoid high-F water to prevent fluorosis[27] by means of choosing alternative low-F water or moving to a low-F community.

Many epidemiologic studies relied on aggregated data (eg, the mean F level in a village's water and the mean IQ of its schoolchildren) to generate hypotheses about potential associations. Using the ecological study design inherently limits the ability to infer causal relationships at the person level due to the ecological fallacy or the mistaken assumption that associations observed between-group means necessarily apply to people within those groups.[124] Taylor and colleagues'[6] mean-effects and dose-response meta-analyses of F and IQ are based on aggregated group-level data, wherein both exposure (ie, the water F level represents mean F, not a person's, exposure) and

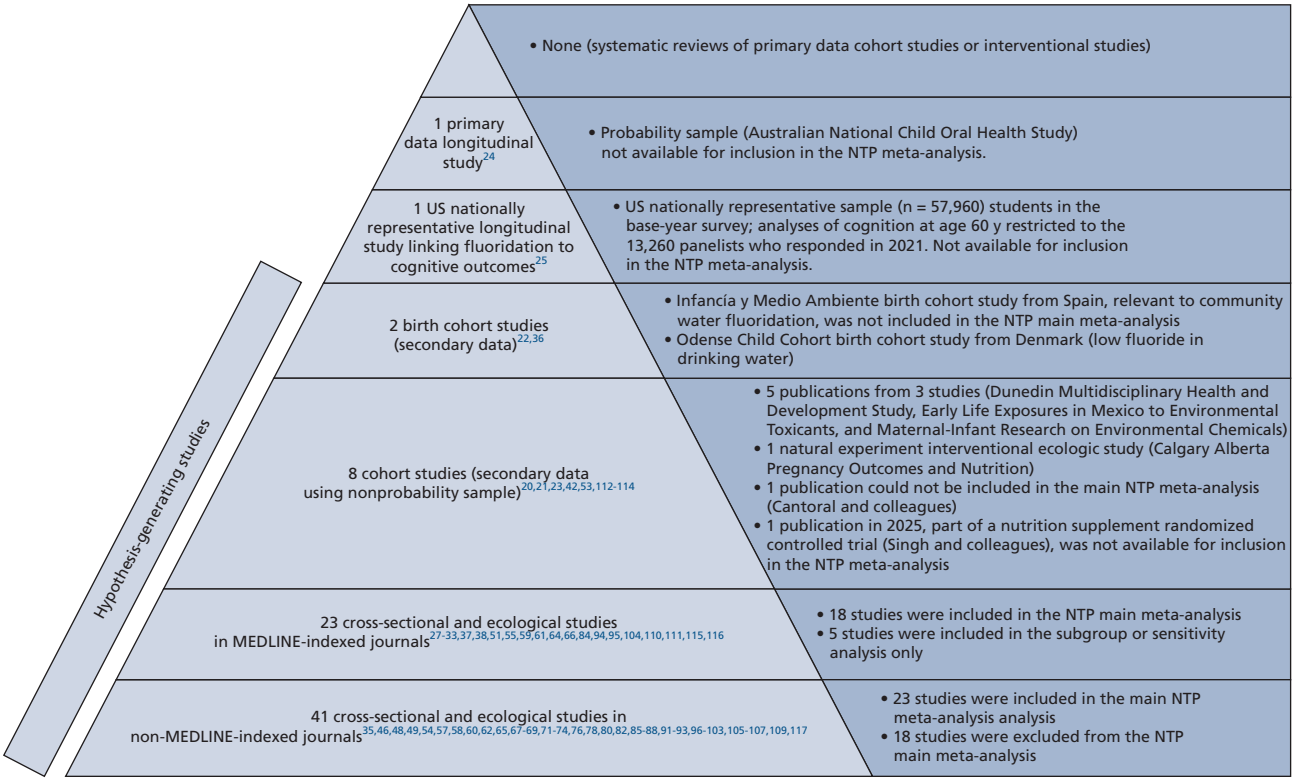


Figure 1. Levels of evidence: classification of fluoride and IQ or cognition epidemiologic studies (Supplemental Material, available online) according to study design. Randomized clinical trials are not feasible because intentionally assigning pregnant women or children to varying levels of fluoride exposure over long periods would be ethically unacceptable and logistically impractical, given the need for long-term cognitive follow-up. Therefore, this category is not included in the evidence pyramid. Taylor and colleagues[6] (eTable 1 in Taylor and colleagues) included 63 of 74 articles in the main meta-analyses (Supplemental Material). A study from Valdez Jiménez and colleagues[118] was excluded because the National Academies of Sciences, Engineering, and Medicine Committee[7] (page 39 of the report) found multiple errors and internal validity issues among its small cohort of 65 participants. Three additional cohort studies, published since December 2024 (Do and colleagues,[24] Singh and colleagues,[113] and Warren and colleagues[25]) were added (Supplemental Material). NTP: National Toxicology Program.

outcome (mean IQ of the group) are aggregated at the study level. Therefore, although Taylor and colleagues'[6] findings may provide a descriptive summary of between-group differences, they cannot be used to establish a causal relationship between F exposure and IQ.

Limitations of Secondary Data-Based Cohort Studies

The cohort studies Taylor and colleagues[6] included relied on analyses of existing data sets that were not originally designed to evaluate neurodevelopmental outcomes in relation to F exposure. However, using secondary data introduces substantial limitations.[125,126] In 2 of the 4 studies, nonprobability sampling was used.[21,34] Several key variables, including F exposure and potential confounding variables (eg, multiple 24-hour urinary F samples, maternal IQ, home environment, socioeconomic status, and environmental coexposures), were either not measured at all or were not measured with sufficient accuracy or frequency to support valid causal inferences. Thus, although secondary cohort analyses can detect associations that can be used to generate hypotheses, in this instance, they do not provide the level of rigor necessary to establish a causal relationship between F exposure and patient IQ outcomes.

Limitations of Exposure Assessment on the Basis of Spot Urinary F Biomarker for Determining Person-Level Total Exposure

According to the WHO, the scientific consensus is that urinary F excretion is not suitable for predicting F intake in people because of significant within- and between-subject variation in urinary F and creatinine excretion rates.[127] Despite this, Taylor and colleagues[6] stated, "Unlike drinking water levels, individual-level urinary fluoride concentrations include all ingested fluoride

and are considered a valid estimate of total fluoride exposure." According to the EPA, this type of urine and serum F biomarker may be considered useless for assessing F neurotoxicity risk on the basis of the National Research Council framework.[2]

Quality Control of Data for Children's IQ Assessment

Many of the challenges highlighted in a National Institute of Environmental Health Services report on neurodevelopmental testing are directly relevant to F-IQ studies and almost certainly contribute to spurious or inconsistent findings.[128] IQ assessments are highly sensitive to cultural, linguistic, and educational differences, making cross-national comparisons problematic when studies are aggregated in meta-analyses. The researchers from the F-IQ studies[20,21,27-30,42-107] in the NTP Monograph rarely discussed the details of test administration, examiner training, blinding, number of examiners, and inter- and intraexaminer reliability; however, these aspects of IQ assessment can introduce substantial, nonrandom errors that distort associations. Guichon and colleagues[129] outlined the implications of using different IQ assessors in each of the cities sampled in the Maternal-Infant Research on Environmental Chemicals study[21,130]; if examiners administer or code the assessments differently, the effect of the examiner cannot be separated from any potential impact of city or F exposure.

Clustering of Observations and Its Impact on Meta-Analysis

In the 2021 NASEM Consensus Study Report, the committee emphasized that many of the F-IQ studies in the NTP Monograph did not adequately account for community-level clustering of observations.[7] Proper control for clustering was often not possible because 1 endemic fluorosis village and 1 nonendemic village were selected per study. In other studies included in the NASEM Consensus Study Report,[7] researchers sampled children from schools within villages or cities (clusters). When multiple children are sampled within the same community, their exposures and outcomes are not statistically independent. This lack of accounting for clustering can lead to underestimated SEs and artificially narrow CIs, potentially biasing inferences about statistical significance. At the meta-analysis stage, such errors compound because a valid synthesis depends on accurate within-study variance estimates.[7]

Covariates and Confounding Variables

All data used in the Taylor and colleagues[6] mean effects meta-analysis and dose-response meta-analyses relied on unadjusted mean IQ scores. Guth and colleagues[11,12] identified the need to assess and control for other contaminants in studies conducted in endemic F areas, as higher concentrations of naturally occurring F (>1.5 mg/L) likely contain metals and metalloids to maintain electroneutrality. Factors such as age, maternal IQ, socioeconomic status, home environment, nutritional status, iodine deficiency, breastfeeding, rural-urban status, alcohol intake, smoking, and the number of IQ testers used in these studies should also be considered.[11,12]

Meta-analyses

The meta-analysis from Taylor and colleagues[6] included 74 F-IQ epidemiologic studies. An assessment indicated that in each study, researchers used either a cross-sectional design or a secondary-data cohort design to generate hypotheses (Supplemental Material). Consequently, none are methodologically appropriate for establishing causal relationships (Figure 1). The 2 studies[24,25] that are ranked highest in Figure 1 were published later and therefore could not be included in Taylor and colleagues[6] meta-analyses. Both Do and colleagues[24] and Warren and colleagues[25] assessed the hypothesis that CWF causes lower IQ scores or cognitive harm and found no evidence of an adverse effect.

Mean-Effects: Difference in Mean IQ Scores Between the Highest and Lowest F Exposure Groups (Group Level)

Taylor and colleagues[6] synthesized data from 59 studies[20,21,27,28,30,31,33,35,37,38,46,48,49,51,54,55,58-62,64,65,67-69,71-74,76,78,80,82,84-88,91-94,96-103,105-107,109-111,131,132] and found a statistically significant relationship between IQ scores and F concentration in water or urine among groups of children living in areas with the highest and lowest F concentrations. To account for differences in IQ tests

and scales, Taylor and colleagues calculated the difference in mean IQ scores between the highest and lowest F-exposed groups and divided it by the pooled SD to obtain the standardized mean difference (SMD).

The pooled estimate of the SMD in IQ between the highest exposed and reference (lowest) groups was −0.45 (95% CI, −0.57 to −0.33; $P < .001$; $I^2 = 94\%$). Results of a subgroup analysis revealed that the association effect size was smaller among those classified as studies with low risk of bias (SMD = −0.19; 95% CI, −0.39 to −0.04; $P = .01$; $I^2 = 87\%$). The reported effect size of 0.19 SMD in the mean-effects meta-analysis for studies with lower risk of bias is considered negligible, whereas the SMD of 0.45 for all studies is considered small.[40]

The SMD between the highest and lowest F-exposed groups varied substantially from −5.34 through 0.43, which was also reflected in the high heterogeneity value (I^2) of 94% across studies. It is inappropriate to use a single summary estimate, as it does not accurately represent the different effects across these studies.[108] Furthermore, there was evidence of publication bias, meaning some studies with null or positive findings were missing; thus, the summary estimate likely overestimated the true effect.[6,15]

Relevance of the Findings to CWF

Figure 2 contains a comparison of mean IQ scores from 5 studies[20-24] conducted in fluoridated and nonfluoridated areas. These results, along with the covariate-adjusted scores in the studies, are consistent with the lack of association between CWF and IQ scores.

Dose-Response Mean-Effects Meta-Analysis (Group Level)

Taylor and colleagues[6] extended the mean-effects meta-analysis to a dose-response analysis by means of including multiple exposure groups in some studies. According to the NASEM Consensus Study Report, "Those analyses fall outside the scope of the NTP monograph, which focuses on hazard identification and not dose–response assessment. ***Given the substantial concern regarding health implications of various fluoride exposures, comments or inferences that are not based on rigorous analyses should be avoided***"[7] (emphasis original). Table 2 presents an analysis of dose-response relevant to CWF (as shown in Table 2 of Taylor and colleagues[6]) to illustrate why this analysis is not credible.

Using the group-level measure of F exposure, Taylor and colleagues[6] found that the pooled changes in SMD from the linear model of the dose-response meta-analyses were statistically significant ($\beta = –0.08$; 95% CI, –0.15 to –0.002), suggesting that the association is detectable even in urinary F concentrations below 1.5 mg/L. However, a closer examination of the data from 4 studies[21,27,30,37] revealed that Taylor and colleagues[6] included children from endemic fluorosis areas, and the mean IQ responses across the 4 studies were inconsistent (Table 2). Moreover, Yu and colleagues' study,[30] the only one in which a decline in IQ was reported, accounted for 69% (2,886/4,179) of the total sample. Results of a piecewise regression analysis in this study revealed a threshold effect, indicating no association between IQ and water F levels below 3.4 mg/L or urinary F levels below 1.6 mg/L. According to Crippa and Orsini,[133] unless all selected studies share a common dose-response model, it is inappropriate to pool the data.

Urinary F and IQ Regression Slopes Meta-Analysis (Person Level)

Taylor and colleagues[6] conducted a meta-analysis of 13 studies[21,27-38] in which researchers reported regression slopes to estimate associations between person-level urinary F exposures and children's IQ. Data from the studies were pooled using a random-effects model. However, not all the regression coefficients were used as reported in the studies; instead, 7 of 13 were calculated.[29,30,32,33,36-38] Antoon and Kumar[39] identified several concerns, including selecting 4 articles[28-30,32] from a study conducted in Tianjin City, using a formula to convert log-transformed data to a linear scale and pooling multiple regression coefficients from nonlinear models.[22]

Taylor and colleagues[6] (as shown in their Table 3) found that, for person-level urinary F measurements, a 1-mg/L increase was associated with a decrease in IQ score of 1.63 points (95% CI, −2.33 to −0.93; $P < .001$; $I^2 = 60\%$), with evidence of substantial heterogeneity. Among studies with low risk of bias, the decrease was 1.14 points (95% CI, –1.68 to –0.61; $P < .001$; $I^2 = 28\%$), indicating low heterogeneity but with publication bias.[29,30,32-34,36,37] Considering that the

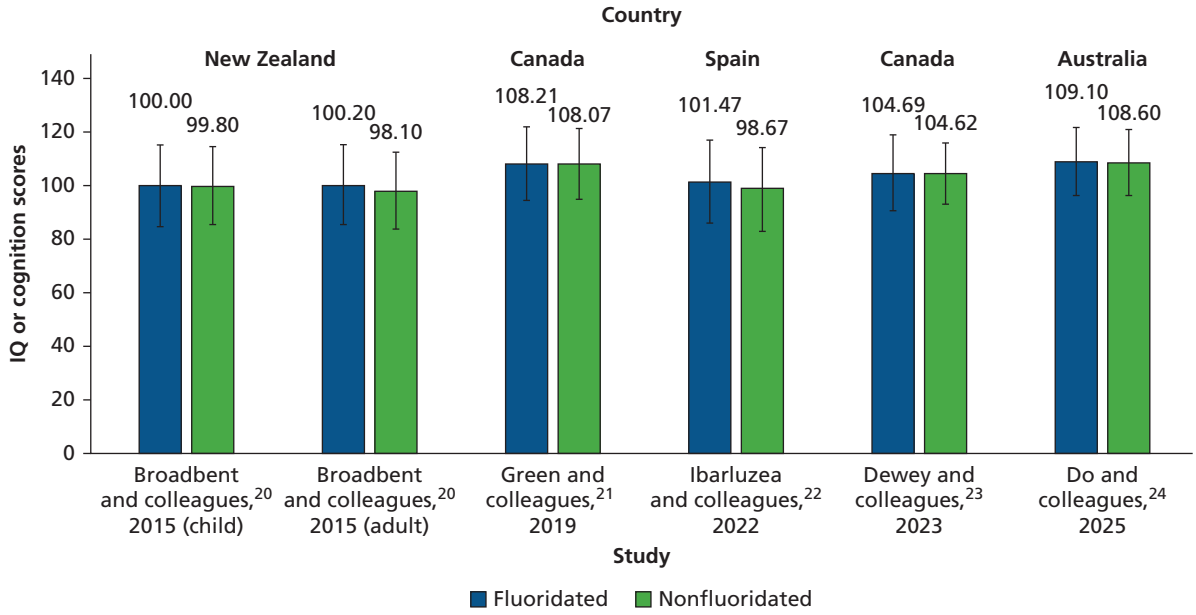


Figure 2. A comparison of mean IQ or cognition scores (unadjusted) from studies conducted in fluoridated and nonfluoridated communities. Data from individual studies were abstracted to compare the IQ scores from fluoridated and nonfluoridated areas.

typical difference in urinary F between fluoridated and nonfluoridated communities is approximately 0.5 mg/L ($\approx$ 0.2 vs 0.7 mg/L), this effect size amounts to approximately one-half an IQ point. Results of a reanalysis by means of subgrouping 5 studies[21,30,36-38] conducted in nonendemic areas (Figure 3) revealed a positive, but not statistically significant, association between F exposure relevant to CWF and IQ (β = 0.27; 95% CI = –0.76 to 1.29).

DISCUSSION

The epidemiologic evidence indicates that F exposure at CWF levels in New Zealand, Canada, Spain, Australia, United Kingdom, and United States (0.7-1.1 mg/L) is not associated with lower IQ scores or cognitive deficits in children.[20,22-25,130] Kumar and colleagues[15] concluded that when only the 8 relevant studies[20-22,42,49,59,78,105] conducted in nonendemic fluorosis areas were considered, the results of the meta-analysis of mean effects clearly showed that CWF F exposure was not associated with lower IQ scores in children. In 2025, Warren and colleagues[25] analyzed a nationally representative cohort of high school and beyond participants (n = 57,960), linking community drinking water F levels in the United States from conception through grade 12 to adolescent test performance and cognition at approximately 60 years of age. Overall, the findings did not show cognitive harm at typical US fluoridation levels and suggest small adolescent benefits. Furthermore, in a study of more than 80,000 military recruits in Sweden, researchers found no effect of F on cognitive ability,[134] and results from a study in Calgary showed that cessation of water fluoridation did not result in increased IQ scores in children.[23] Savitz[135] summarized the evidence best by means of asserting, "Until clear evidence exists that water fluoridation lacks public health benefit or compelling evidence of harm at the level of fluoride exposure in fluoridated water, neither of which has occurred, it seems foolhardy to interfere with a long-established and well-recognized public health success."

Although the association between F exposure at concentrations exceeding 1.5 mg/L in drinking water and child IQ has received considerable attention, the NASEM Consensus Study Report committee rejected NTP's presumed neurodevelopmental hazard determination.[7] The EFSA Scientific Committee[10] concluded that evidence from cross-sectional studies of high F levels suggests a link between total F intake and adverse effects on brain development at drinking water concentrations exceeding 1.5 mg/L. The committee observed that on their own, however, these cross-sectional studies[30,38,44-46,59] are suggestive but not sufficiently robust as a stand-alone line of

Table 2. Results of studies with low risk of bias included in Taylor and colleagues[6,*] dose-response mean-effects meta-analysis using group-level measures of urinary F exposure (mean, < 1.5 mg/L) with heterogeneous IQ response among the 4 studies.

STUDY	F CONCENTRATION IN WATER AND URINE	GROUP LEVELS (DEFINED AS DENTAL FLUOROSIS OR WATER F)	NO.	URINARY F, MG/L	IQ, MEAN (SD)	RELATIONSHIP BETWEEN URINARY F AND IQ RESPONSE
Ding and Colleagues,[27] 2011	"Most regions in Hulunbuir City, Inner Mongolia, are identified as drinking water type of fluorosis areas." Water F (4 sites, 16 samples) ranged from 0.24-2.84 mg/L. Mean (SD), 1.31 (1.05) mg/L Urine F concentrations at these 4 sites were 0.100-0.552 mg/L (as the control group), 0.554-0.810 mg/L (as the low-exposure group), 0.818-1.301 mg/L (as the medium-exposure group), and 1.317-3.550 mg/L (as the high-exposure group). However, the mean IQ score was not provided. Dental fluorosis conditions were the dose categories.	Normal teeth	136	Mean, 0.80 (range, 0.1-3.44)	104.07 (12.30)	The data indicate an inconsistent relationship, suggesting children with mild dental fluorosis had the highest mean IQ score. Did not account for cluster sampling design. SMD between moderate dental fluorosis vs normal dental fluorosis, 0.04 (95% CI, –0.45 to 0.36), was not statistically significant
		Questionable fluorosis	54	Mean, 1.13 (range, 0.19-3.41)	103.00 (16.10)	
		Very mild fluorosis	74	Mean, 1.11 (range, 0.16-3.55)	102.11 (15.05)	
		Mild fluorosis	39	Mean, 1.31 (range, 0.25-3.32)	106.03 (12.33)	
		Moderate fluorosis	28	Mean, 1.46 (range, 0.22-3.22)	103.54 (13.59)	
Yu and Colleagues,[20] 2018	F in water (mg/L): Lower (≤ 1) Mean (SD), 0.50 (0.27); range, 0.2-1 mg/L Higher (> 1) Mean (SD), 2.00 (0.75); range, 1.10-3.90 mg/L F in urine (mg/L): Range, 0.01-5.54 Low: mean (SD), 0.41 (0.49) High: mean (SD), 1.37 (1.08)	Lower F	1,636	Mean, 0.41	107.40 (13.00)	Decline in IQ; SMD is negative, –0.08 (95% CI, –0.15 to –0.01) and statistically significant Did not account for complex survey design No effect < 1.5 mg/L F either in water or urine Adjusted regression coefficient Water F: 0.20-3.40 mg/L $\beta = -0.04$; 95% CI, −0.33 to 0.24), not statistically significant Water F: 3.40-3.90 mg/L −4.29 (95% CI, –8.09 to –0.48), statistically significant Urinary F 0.01-1.60 mg/L (low exposure relevant to community water fluoridation) 0.36 (95% CI, –0.29 to 1.01) per 0.5 mg/L, not statistically significant 1.60-2.50 mg/L (medium exposure) –2.67 (95% CI, −4.67 to −0.68) per 0.5 mg/L, statistically significant 2.50–5.54 mg/L (high exposure) −0.84 (95% CI, −2.18 to 0.50) per 0.5 mg/L, not statistically significant
		Higher F	1,250	Mean, 1.37	106.40 (12.30)	
Green and Colleagues,[21] 2019 .	Water F: Lower F (nonfluoridated mean [SD], 0.13 [0.06]) Higher fluoride (fluoridated mean [SD], 0.59 [0.08])	Lower F	238	Mean, 0.40	108.07 (13.31)	No difference in IQ. Did not account for cluster sampling design. SMD is positive, 0.01 (95% CI, −0.19 to 0.21), not statistically significant
		Higher F	162	Mean, 0.69	108.21 (13.72)	
Lin and Colleagues,[37] 2023	The public water supply systems in Taiwan, which serve approximately 95% of the population, are not fluoridated and have a maximum F limit of 0.8 mg/L. Taiwan started the salt fluoridation program in 2016. Dental fluorosis conditions were the dose categories.	No dental fluorosis	429	Mean, 0.38	97.72 (15.09)	Increase in IQ was not statistically significant SMD is positive, 0.10 (95% CI, −0.10 to 0.29), and was not statistically significant
		Dental fluorosis	133	Mean, 0.46	99.22 (15.87)	

* Taylor and colleagues[6] pooled changes in standardized mean differences (SMDs) from the linear model from the dose-response mean-effects meta-analysis using group-level measures of urinary fluoride (F) exposure (< 1.5 mg/L), 4 studies with low risk of bias. Mean water F values ranged from 0.13 to 2.84 mg/L; 7 effect estimates; n = 4,179; mean urinary F, < 1.5 mg/L; parameter estimates: $\beta = -0.08$ (95% CI, –0.15 to –0.002); $P = .04$.

evidence to derive a health-based guidance value due to the lack of clarity about study participants' characteristics, including parental education and socioeconomic status, inadequate control of confounding variables, and missing information on the duration and timing of F exposure.[10] In several cross-sectional studies,[27,30,38,44-46] researchers recruited participants who were long-term

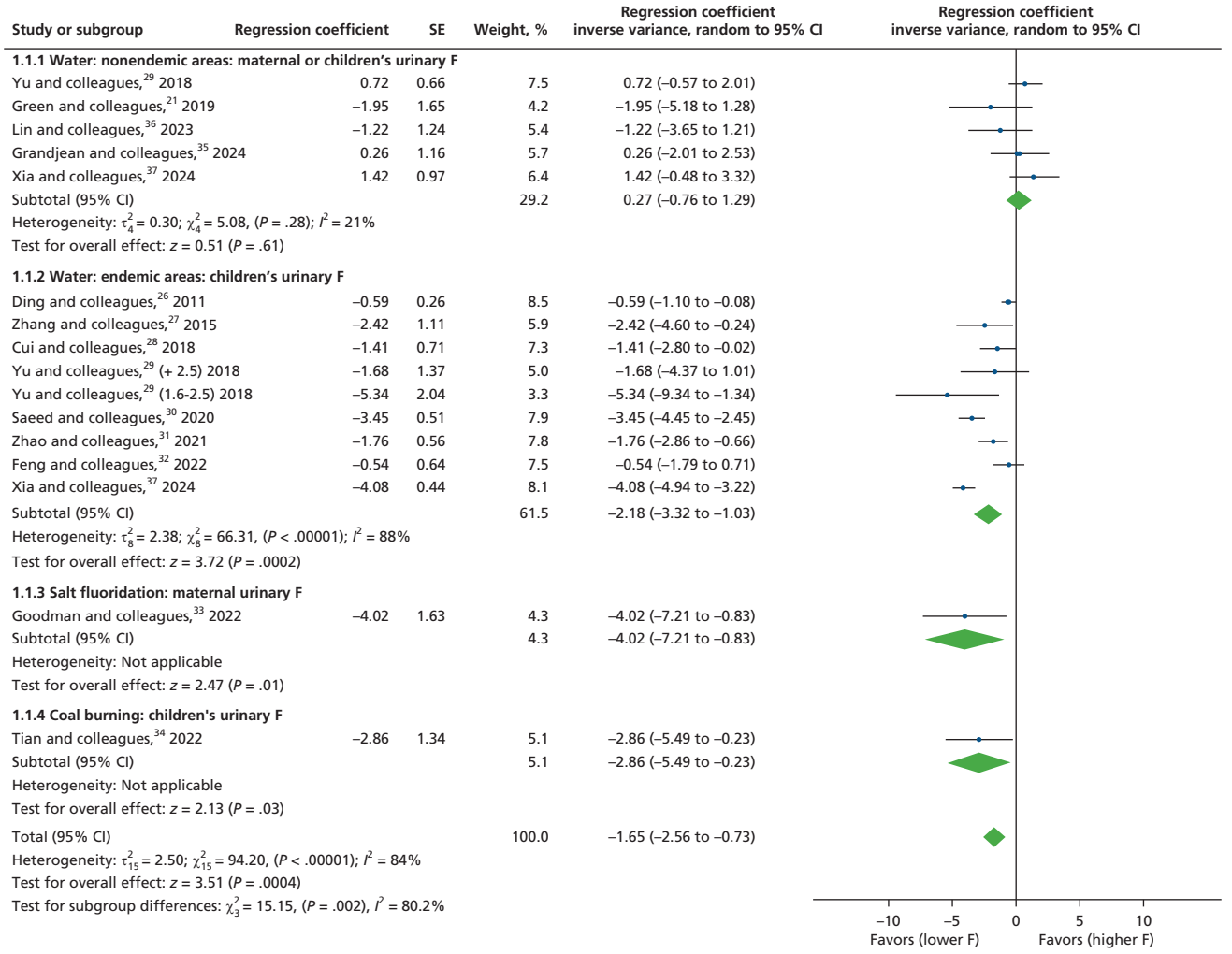


Figure 3. Forest plot for random-effects meta-analysis of regression slopes of the association between person-level urinary fluoride (F) measures and IQ scores in children according to subgroups. The effect measures are regression slopes, β per 1 mg/L increase in spot urinary F. The βs for studies obtained from the original article are indicated with boxes representing the weights, and the pooled estimate is indicated with a diamond. The error bars represent 95% CIs for the study-specific βs. The reported CI for the Xia and colleagues[38] study has an error in the article ($\beta = -4.08$; 95% CI, –3.04 to –1.32) for the high-F group. For this analysis, we estimated the 95% CI for Xia and colleagues ($\beta = -4.08$; 95% CI, –4.94 to –3.22) to reproduce Taylor and colleagues'[6] pooled estimate ($\beta_{pooled} = -1.63$; 95% CI, –2.33 to –0.93; $P < .001$; $I^2 = 60\%$). This may have resulted in a slight difference in the pooled estimate ($\beta_{pooled} = -1.65$; 95% CI, –2.56 to –0.73; $I^2 = 84\%$).

residents in areas with low or high F concentrations in drinking water; however, IQ may have influenced F intake in endemic areas due to selective adherence to guidance to avoid high F water. Other limitations not cited by the EFSA Scientific Committee include the lack of accounting for the clustering of observations, the small effect sizes reported in meta-analyses characterized by high heterogeneity, the absence of support from animal studies to establish a mechanism for how F reduces IQ, the inadequacy of the data to undertake dose-response modeling, and evidence of publication bias. These limitations greatly reduce the ability to use the data for assessing the causal role of F in children's IQ deficits.

In recommending the 1.5 mg F/L as the guideline value for drinking water, which was set to prevent dental and skeletal fluorosis, the WHO[136] noted that it is essential to consider the mean daily intake of water and of F from other sources (eg, from food and air) by the population of interest. Regarding F-IQ studies in endemic areas[46,48,49,57,59,60,62,72,76,80,88,93,94,98,100,105-107] in countries such as China and India, Kumar and colleagues[15] estimated the mean F concentration in water in the highest-exposed group was approximately 3.7 mg F/L. Li and colleagues[137] estimated

the total mean daily F intake in China to be approximately 7.85 mg and 14.13 mg within the exposure range of 2.62 through 3.56 mg F/L and from 4.32 through 7.97 mg F/L in water, respectively, on the basis of the mean daily intake of the population. This mean F intake in endemic areas of China is 3 times higher than the estimated daily F intake of 2.49 mg for adults drinking 2 L (16 cups) of water per day in a fluoridated community in the United States.[138]

The EPA uses a rigorous risk assessment framework that necessitates not only hazard identification but also dose-response modeling, exposure assessment, and integration of evidence from human, animal, and mechanistic studies to support regulatory standards.[139] Kumar and colleagues[140] found that a data quality assessment revealed serious flaws that render the maternal urinary studies[21,22,34,36] unacceptable for hazard assessment and benchmark dose modeling. In its 2010 dose-response assessment, the EPA[141] identified 1.87 mg/L of F in drinking water (a reference dose of 0.08 mg/kg/d) as a protective benchmark for severe dental fluorosis on the basis of results of research showing that this level is associated with a low prevalence (approximately 0.5%) of severe fluorosis cases in the United States.

CONCLUSIONS

The available evidence from articles on F and cognitive abilities, specifically those assessing CWF F concentrations, clearly show that CWF is not associated with children's IQ deficits. Although there may be an association between naturally occurring F at high levels (> 1.5 mg/L) and lower IQ in some countries outside the United States, cross-sectional and ecological studies conducted in these regions have been unable to establish that higher levels of F cause IQ deficits due to numerous methodological shortcomings. Dental, medical, and public health care practitioners should use this evidence to inform, educate, and empower decision makers and the public regarding the safety and benefits of CWF. ■

DISCLOSURES

Dr. Kumar and Dr. Levy are members of the American Dental Association's National Fluoridation Advisory Committee. Dr. Kumar was a reviewer of the National Academies of Sciences, Engineering, and Medicine report and Review of the Revised NTP Monograph on the Systematic Review of Fluoride Exposure and Neurodevelopmental and Cognitive Health Effects: A Letter Report (2021).

SUPPLEMENTAL DATA

Supplemental data related to this article can be found at: https://doi.org/10.1016/j.adaj.2026.02.010.

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